

BAAZ JKBOSE

🔥 2026 CLASS 11th 🔥

**UNIT AND
MEASUREMENTS**

PHYSICS

LECTURE-2

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TOPICS TO BE COVERED

Division

1 Unit

2 System of Unit

3 Dimensions

4 Application of Dimension

Questions

①

How many fundamental physical quantities are in Physics?

1 5

2 7 ✓

3 2

4 9

Questions

②

Which of following is a fundamental physical quantity?

- 1 Angle ✗
- 2 Charge ✗
- 3 Temperature ✓
- 4 Density ✗

Questions

③

The base quantity among the following is-

1 Speed ✗

2 Weight ✗

3 Length ✓

4 Area ✗

Questions

④

Which of the following is not the name of a physical quantity:-

1 ✓ Kilogram

2 impulse ✗

3 energy ✗

4 density ✗

unit of mass PQ

Questions

⑤

~> (class 10th)

Ampere second is a unit of_____.

1 Current

2 Charge

3 Energy

4 Power

Formula

master
hud

Formula

def

class - 10th

$$I = \frac{Q}{t}$$

$$I \times t = Q$$

$$\text{Ampere} \times \text{Sec} = Q$$

charge

Questions

⑥ ? DPP-①

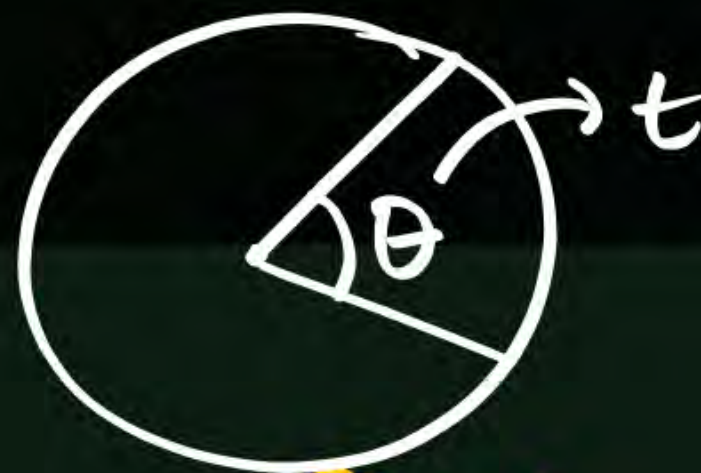


rad/s is the unit of-

Chapter ⑥



$$\text{Angular velocity} = \frac{\theta}{t}$$



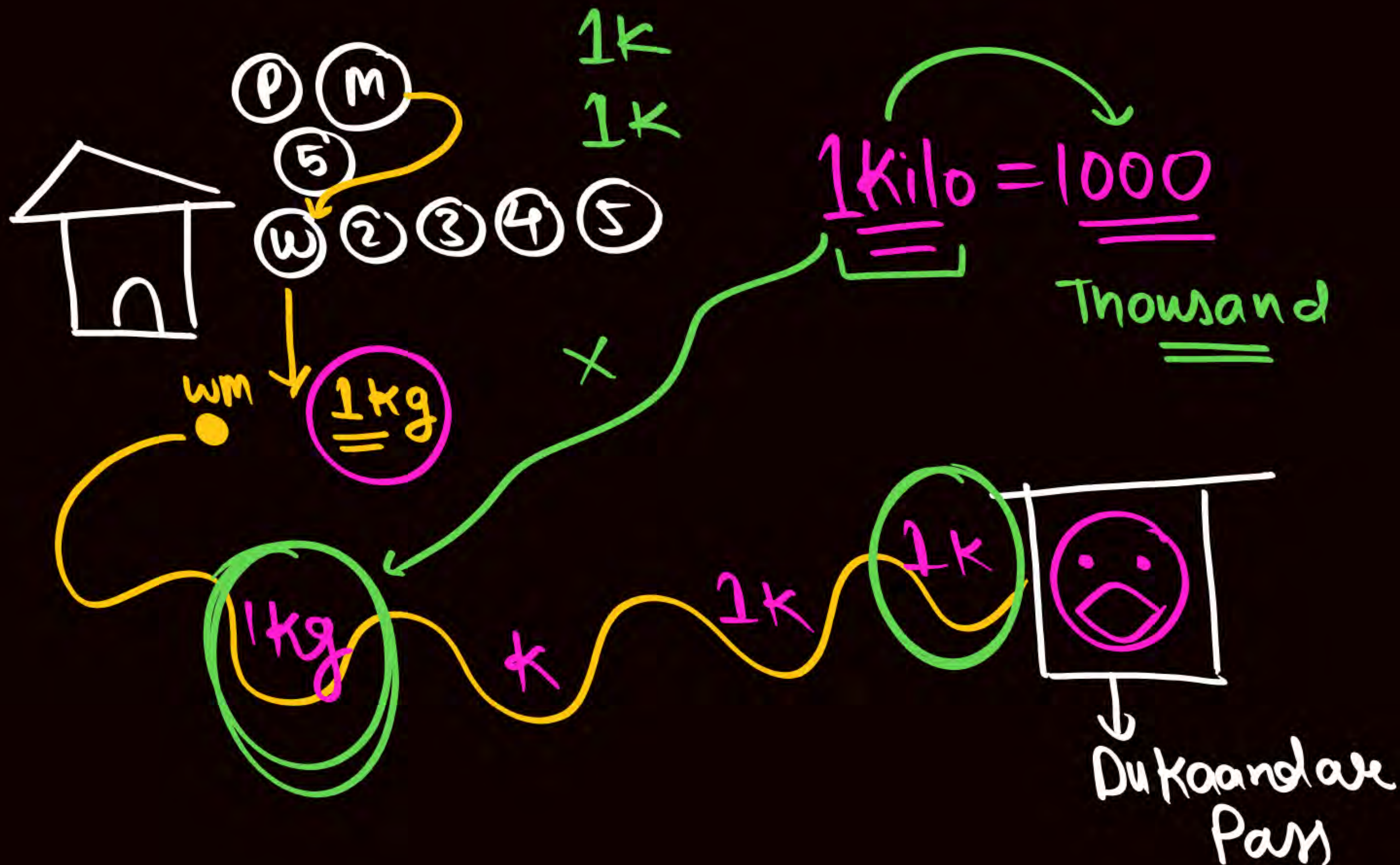
$$= \frac{\text{rad}}{\text{sec}}$$

1 Angular displacement

2 Angular acceleration

3 ✓ Angular velocity

4 Velocity



$$10^6 = 1000000$$

}

106

PYQ 2021



number name

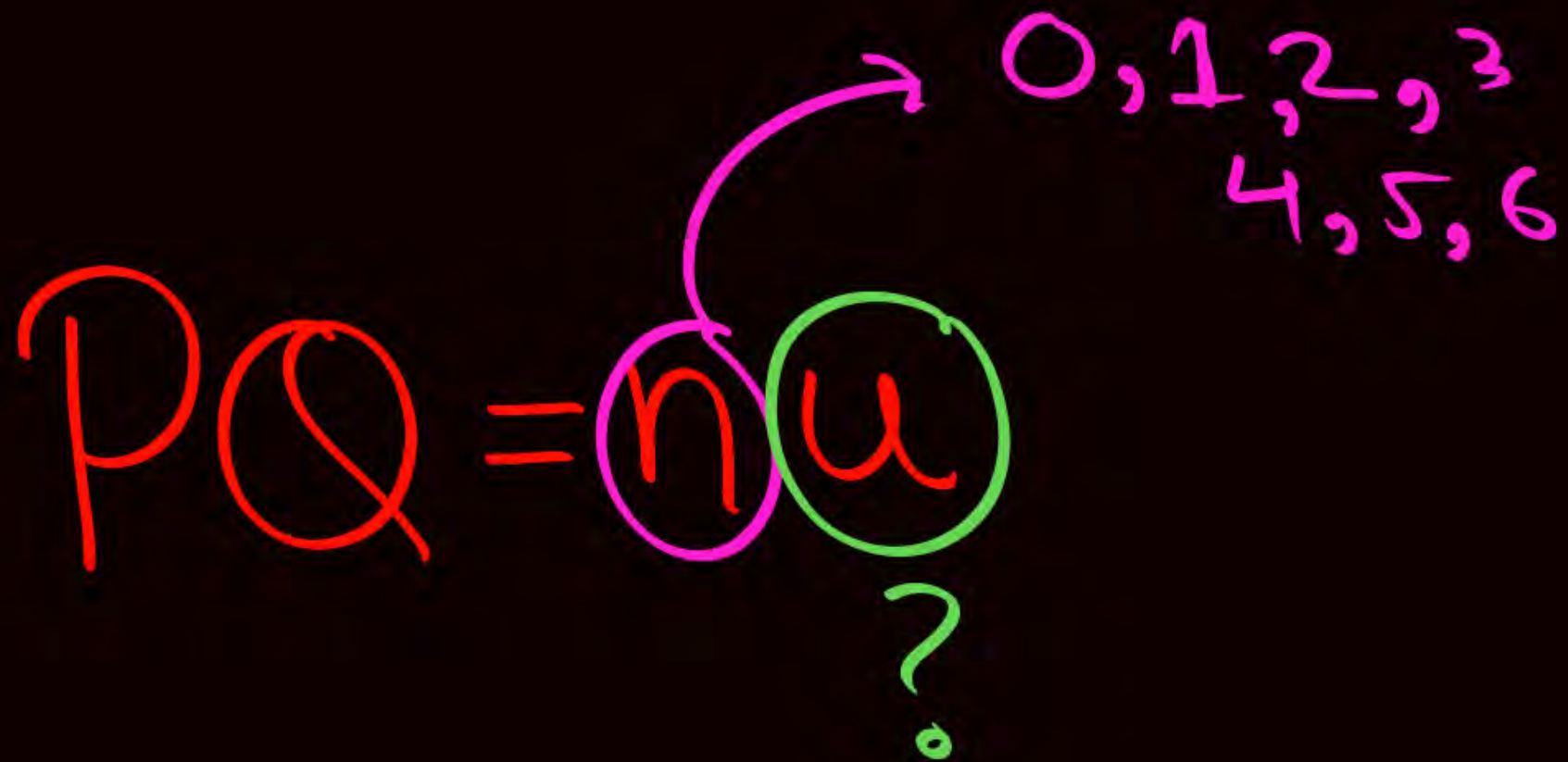
1 Angstrom = 10^{-10}

$1 \text{ \AA} = 10^{-10}$

Chemistry
master

PQ = n u
?

0, 1, 2, 3
4, 5, 6



Unit

The unit is the fixed quantity used to measured the PQ.

Mass →

1kg

Length = 1metre

✓ Ahloo = 2kg

Ribbon = 20metre

✓ Wife = 60kg

good friend = 45kg
(Ashraf)

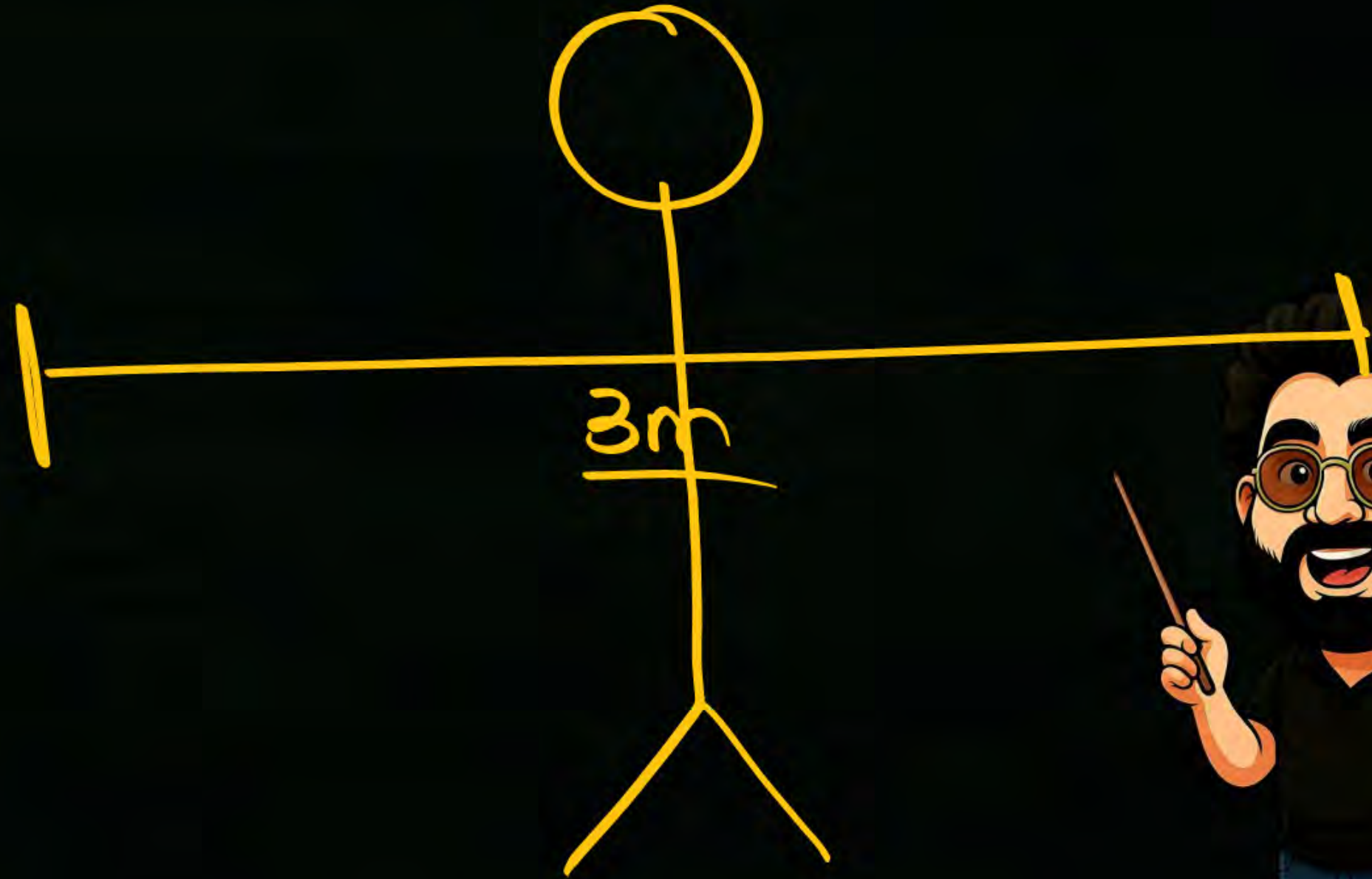
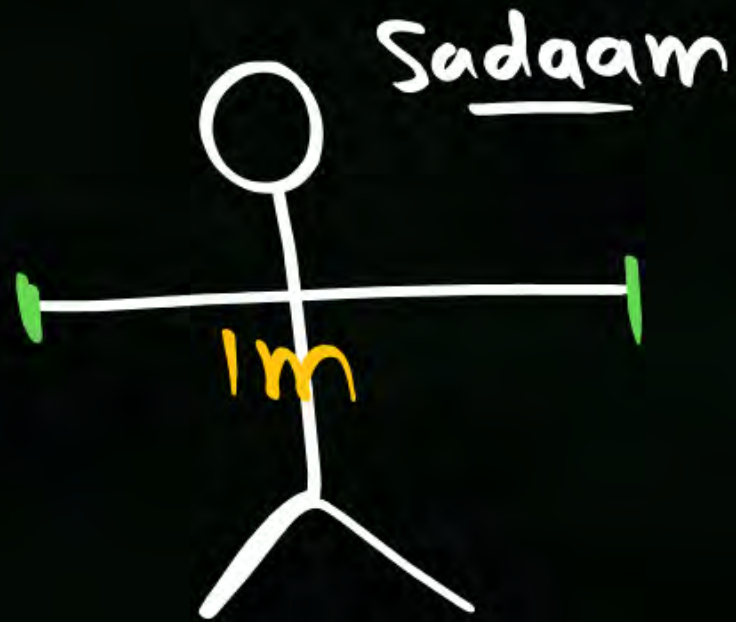


Some old units

→ Non-standard unit =

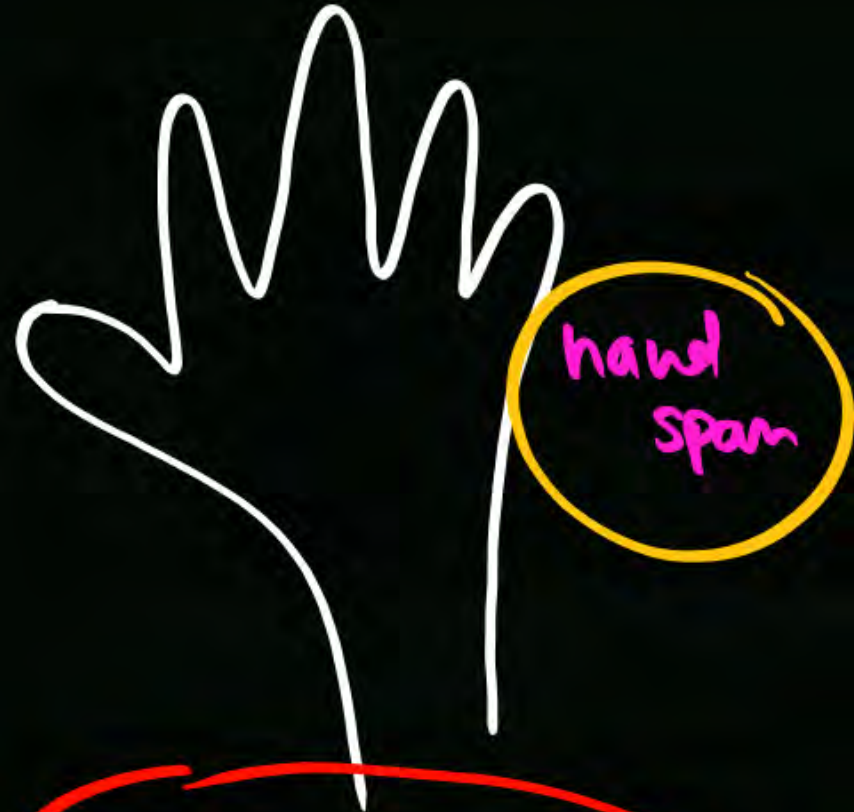
2021

● Arm span



Some old units

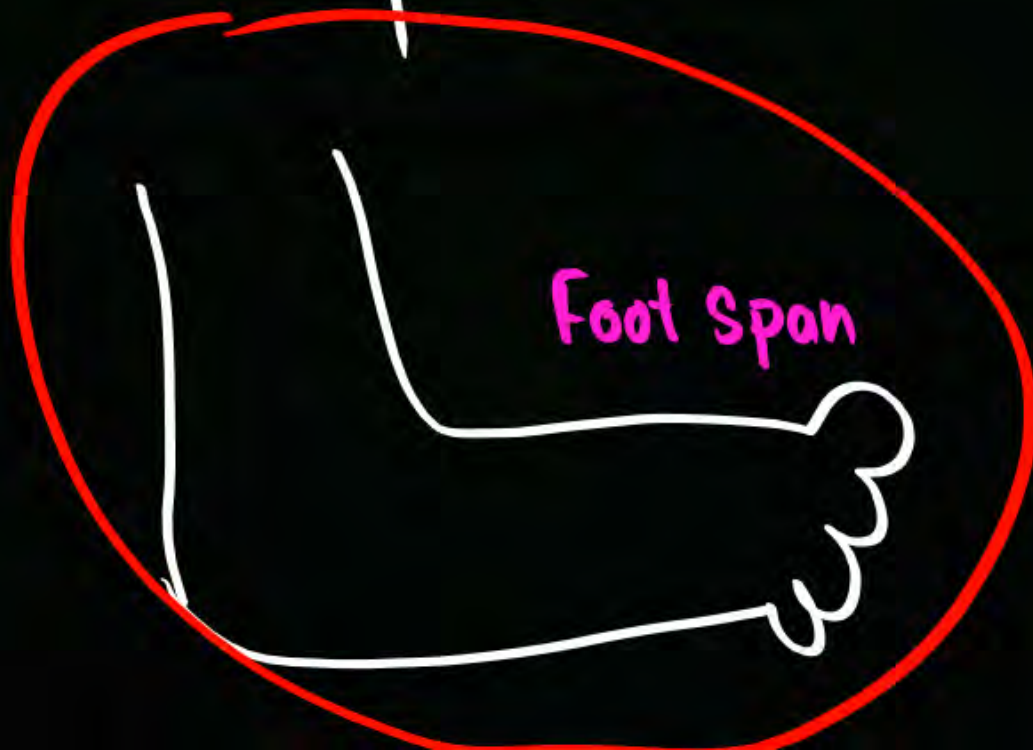
②



③



④



Properties of Good Unit

- It does not change place to Place ✓
- Easy to Understand ✓
- Easy to Handle



System of Unit

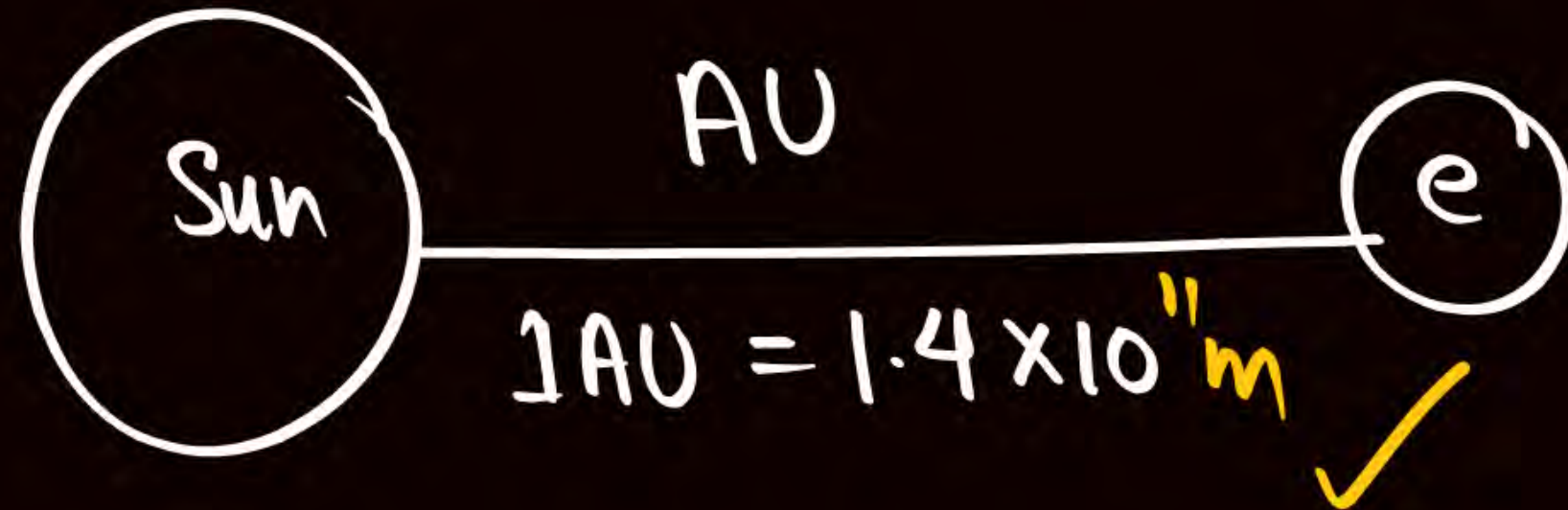
Earlier in different parts of the world different system of units were used for measurement. ✓

System of units	Length	Mass	Time
✓ FPS.	Foot	Pound	Second
C.G.S.	centimetre	gram	Second
S.I System of unit. M.K.S.	metre	Kilogram	Second



Q what is AU (Astronomical unit) pyq 2024 2021

sol AU is the distance b/w earth & Sun



Questions

$$1 \text{ AU} = 1.4 \times 10^{11} \text{ m}$$

AU is the unit of _____.

- 1** astronomy unit
- 2** astronomical unit
- 3** astrological unit
- 4** archaeological unit

? What is light Year (PYQ-2021)

Ans light Year is unit of distance **

→ vacuum • the distance travel by light in one Year. ✓

1 Year

light → ————— →

1 light Year = 9.4×10^{15} m ✓

Questions

Which of the following is not a unit of time? ✓

1 Second

2 Minute

3 Hour

4 ✓ Light year

Ratta (2019, 2021)

Note :- parsec is the unit of distance metre

$$1\text{m} = 100\text{cm}$$

$$\underline{1\text{parsec}} = \underline{3.2} \text{ ly} \quad \begin{array}{c} \text{cm} \\ \hline \text{AU} \\ \hline \text{ly} \end{array}$$

Dimensions

Newton chacha

* old unit

The dimensions are the powers raised to base PQ to determine the relationship b/w basic PQ and derived PQ.

Mass → M

Length → L

Time → T

Current → A

Temperature → K/θ

Luminous intensity → cd

No. of Substance → mol



[Examples]

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

Dimensional Equation

Speed

$$= \frac{L}{T}$$

Dimensions

Speed

$$= \frac{L}{T}$$

Dimensional

formula

② Velocity = $\frac{\text{displacement}}{\text{time}}$

Dimensional equation

= $\frac{L}{T}$

Dimension

Velocity =

$L^1 T^{-1}$

Dimensional formula.

③ acceleration

$$a = \frac{v}{t} \rightarrow L T^{-1}$$

$$a = \frac{L T^{-1}}{T}$$

$$a = L T^{-1} T^{-1}$$

$$a = L \overset{+1}{\underset{-1}{I}} \overset{-2}{\underset{-1}{I}}$$

$$a = \boxed{L T^{-2}}$$

Important question

- Give the dimensional formula of ϵ_0 ?



Important question

- Give the dimensional formula of η ?



Application of Dimension

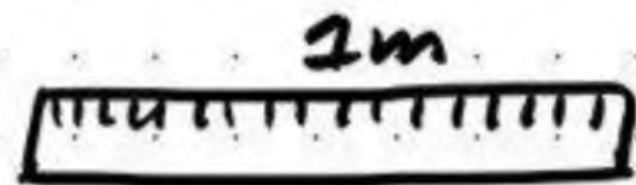
(i) To find out the dimensional formula of PQ .

To find out the dimensional formula of G .



④

unit :- A unit is the fixed quantity in reference to which all other physical quantity are measure



1sec

System of unit

Length

Mass

Time

FPS

Foot

Pound

second.

CGS

centimetre

gram

second

MKS

metre

Kilogram

second

Dimension:- OLD units

The Dimension are the power raised to the base Quantity to determine the Relationship b/w the Derived PQ and the Base physical Quantity

Mass	M
length	L
time	T
current	A
temperature	K
luminous intensity	cd
no- of substance	mole



Yaad
Hogaya

① $\text{Speed} = \frac{\text{distance}}{\text{time}} = \frac{L}{T} = L^1 T^{-1}$

$\text{Speed} = L^1 T^{-1}$

$\text{Speed} = 1 \times L^1 T^{-1}$

$\text{Speed} = M^0 L^1 T^{-1}$

Dimensional
Equation

Dimension

Dimensional formula

⑤ Velocity

$\text{Velocity} = \frac{\text{displacement}}{\text{time}} = \frac{L}{T} = L^1 T^{-1} = M^0 L^1 T^{-1}$

③ Acceleration

$\text{acc} = \frac{v}{T} = \frac{L^1 T^{-1}}{T} = L^1 T^{-1} T^{-1} = L^1 T^{-2} = M^0 L^1 T^{-2}$

④ Force

$F = ma = M L T^{-2}$

⑤ Work

$W = Fd = M L T^{-2} L = M L^2 T^{-2}$

⑥ Potential Energy

$a = g = L T^{-2}$

$P.E = mgh = M L T^{-2} L = M L^2 T^{-2}$

⑦ Kinetic Energy

$K.E = \frac{1}{2} mv^2 = mv^2 = M (L T^{-1})^2 = M L^2 T^{-2}$

⑧ Power

$P = \frac{W}{T} = \frac{M L^2 T^{-2}}{T} = M L^2 T^{-2} T^{-1} = M L^2 T^{-3}$

⑨ Area

$$A = L \times B = L' \times L' = L^2$$

⑩ Volume

$$V = L \times B \times h = L' \times L' \times L' = L^3$$

⑪ Density ρ

$$\rho = \frac{M}{V} = \frac{M}{L^3} = ML^{-3}$$

⑫ Angle θ

$$\theta = \frac{\text{Arc}}{R} = \frac{L}{L} = 1 \text{ (Dimensionless Quantity)}$$

Application of Dimension-

① To find the Dimensional Formula of Physical Quantity Dimensional formula of G

$$F = G \frac{m_1 m_2}{r^2}$$

$$MLT^{-2} = G \frac{MM}{L^2}$$

$$MLT^{-2} = \frac{GM^2}{L^2}$$

$$ML^2T^{-2} = GM^2$$

$$GM^2 = ML^3T^{-2}$$

$$G = \frac{ML^3T^{-2}}{M^2}$$

$$G = \frac{L^3T^{-2}}{M}$$

$$G = M^{-1}L^3T^{-2}$$

$$G = M^{-1}T^{-2}L^3$$

~~4/12~~

Q Find out the Dimensional formula
of Force

Sol

$$F = ma$$

Summary

Dimension

...

class - Ist Division

Number name

A^0

unit

old unit

AU

ly

Parsec

Thank
You